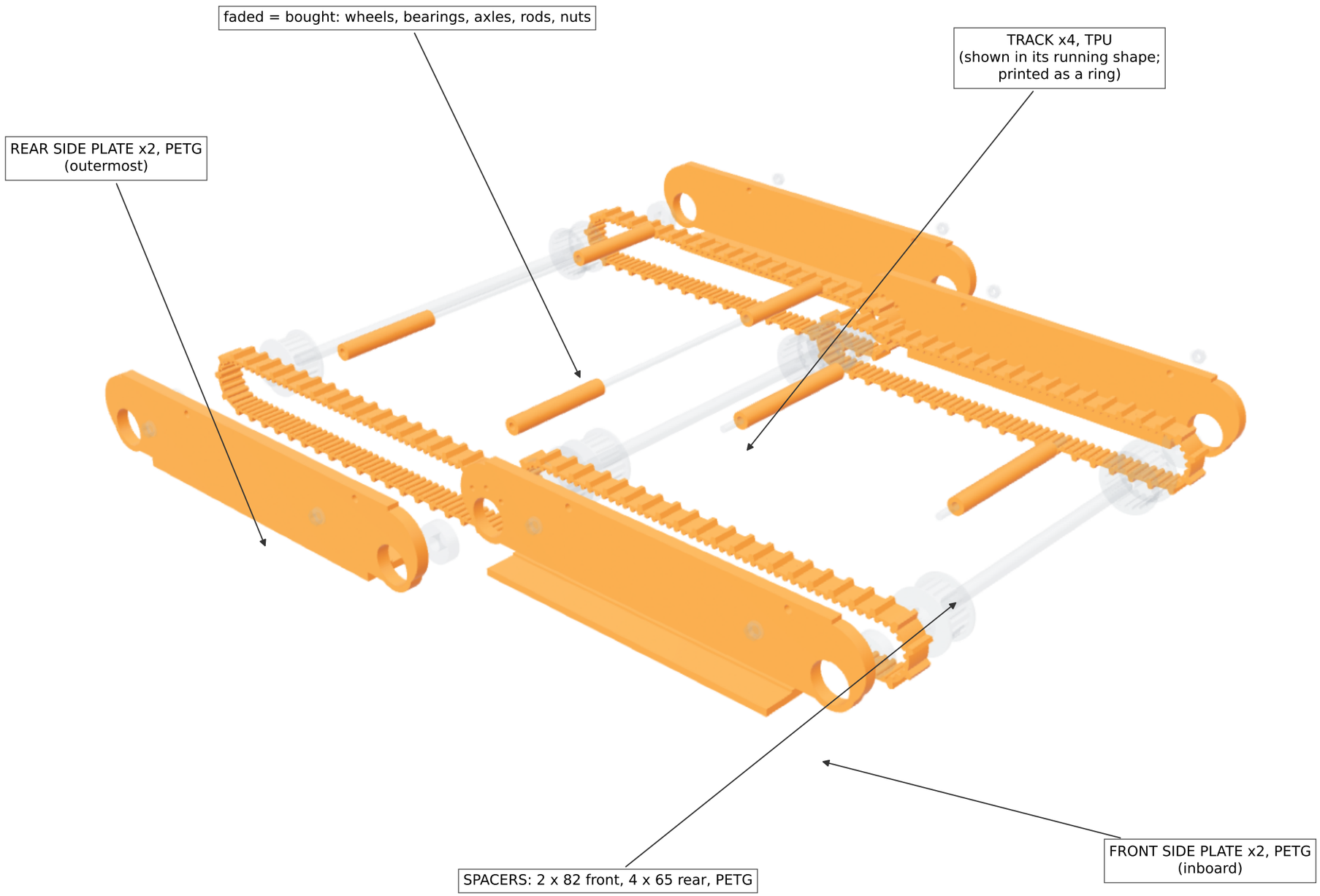
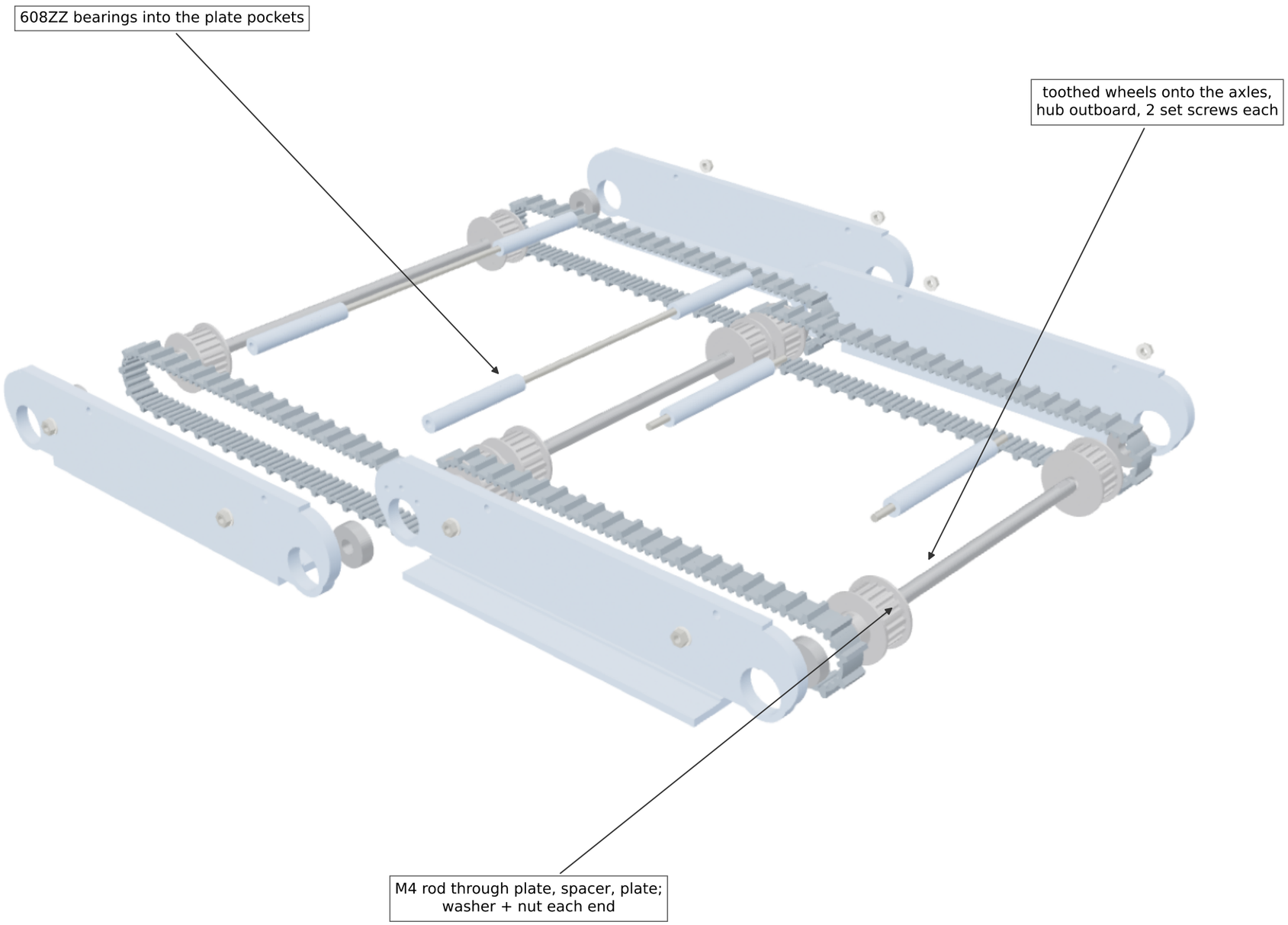
Exploded view: the parts you print (orange), pulled apart along the axles



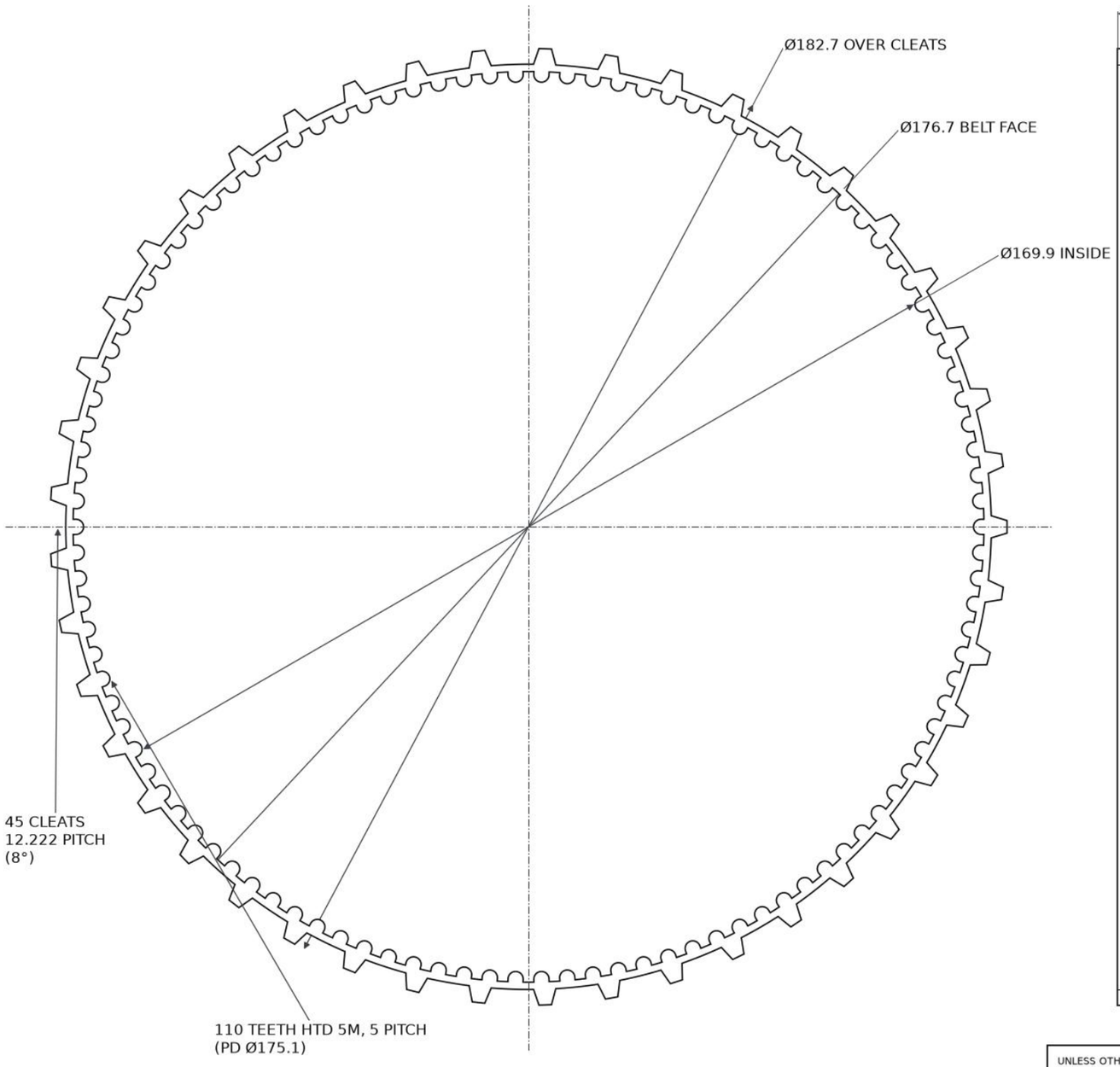
Printed parts: 4 tracks, 4 plates, 6 spacers. Settings and files on the sheets that follow. Rear box / profile frame omitted for clarity.

Exploded view: everything, same spread



Assembly order: bearings into plates; wheels onto axles; one plate on; track over its two wheels; second plate on; cross bars; nuts.

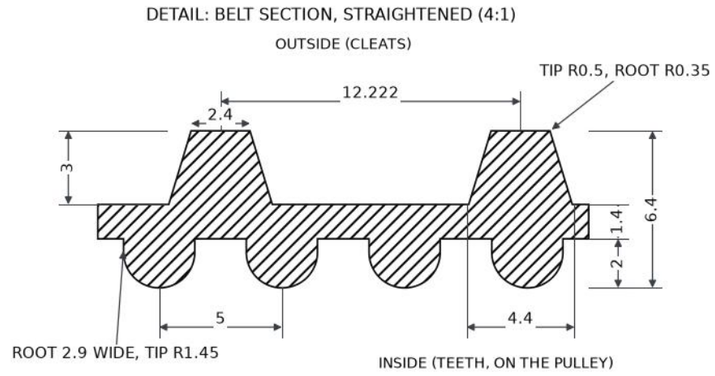
FRONT VIEW



EDGE VIEW



DETAIL: BELT SECTION, STRAIGHTENED (4:1)



NOTES:
1. FILE stl/track_TPU_x4.stl. PRINT STANDING AS A RING, NO SUPPORTS, 0.2 LAYERS, 100% INFILL, 20-30 MM/S, DIRECT-DRIVE EXTRUDER.
2. LOOP 550 ON THE PITCH LINE = 110 TEETH x 5 PITCH; RUNS ON TWO HTD 5M 20T PULLEYS (TS-06) AT 225 CENTRES. ABOUT 32 G.
3. 45 CLEATS AT 12.222 PITCH: 5 CLEATS = 61.11, ONE ES1 STEP ALONG THE SLOPE (35 x 50 AT 1:5), SO EVERY STEP EDGE SITS BETWEEN TWO CLEATS.
4. BED SMALLER THAN 190 MM: ASK FOR THE TWO-PIECE TRACK. PRINT THE 5 MM FIT-TEST SLICE FIRST (stl/fit_test_track_5mm_TPU.stl).

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
TRACK, 110T HTD 5M, 45 CLEATS

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: TPU 95A, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-01

QTY.: 4

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

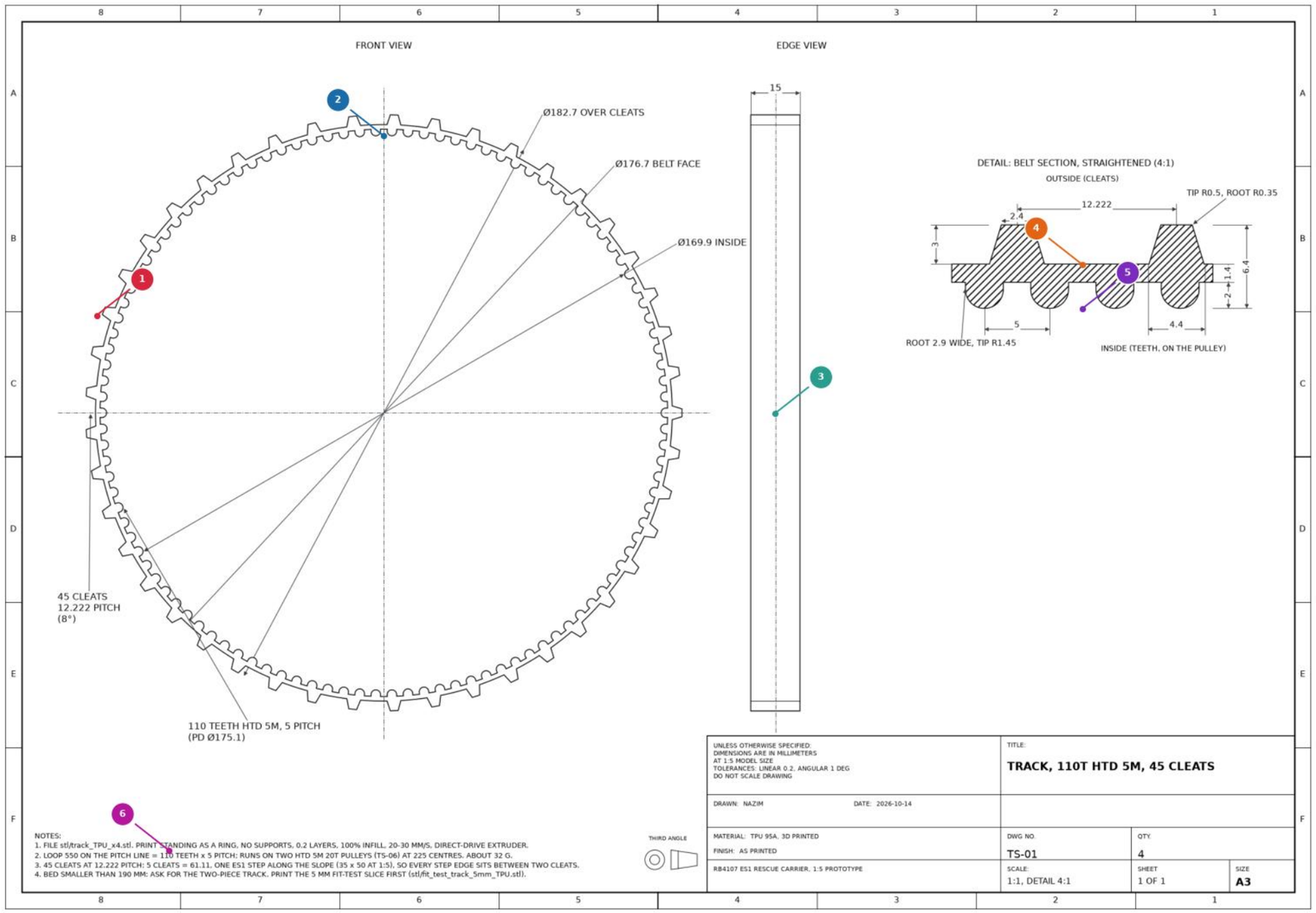
SCALE: 1:1, DETAIL 4:1

SHEET: 1 OF 1

SIZE: **A3**

How to read TS-01: the track

Markers below sit on a reduced copy of sheet TS-01 (previous page, full size). Numbers match the key on the right.



1 Cleats, outside

45 bumps 12.22 apart (8 degrees). Five of them span one ES1 step, so every step edge lands between two cleats and the track holds like a gear on a rack.

2 Teeth, inside

110 HTD 5M teeth, 5 apart. These mesh with the 20-tooth wheels; this is what stops the track slipping.

3 Edge view

The ring seen from the side: 15 wide, 182.7 tall over the cleats.

4 Cleat profile (4:1)

A cleat is 3 tall, 4.4 wide at its base and 2.4 at the tip, with rounded corners so it releases from the step.

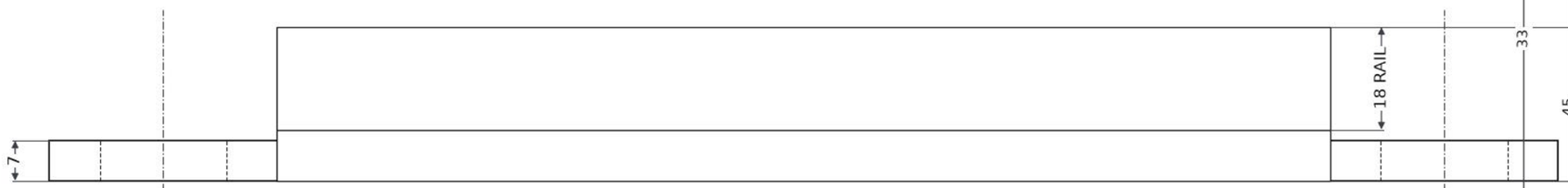
5 Tooth profile (4:1)

A tooth is 2 deep with a 1.45 radius tip and a 2.9 root; the 1.4 backing between teeth and cleats is the belt itself. Whole section 6.4 thick.

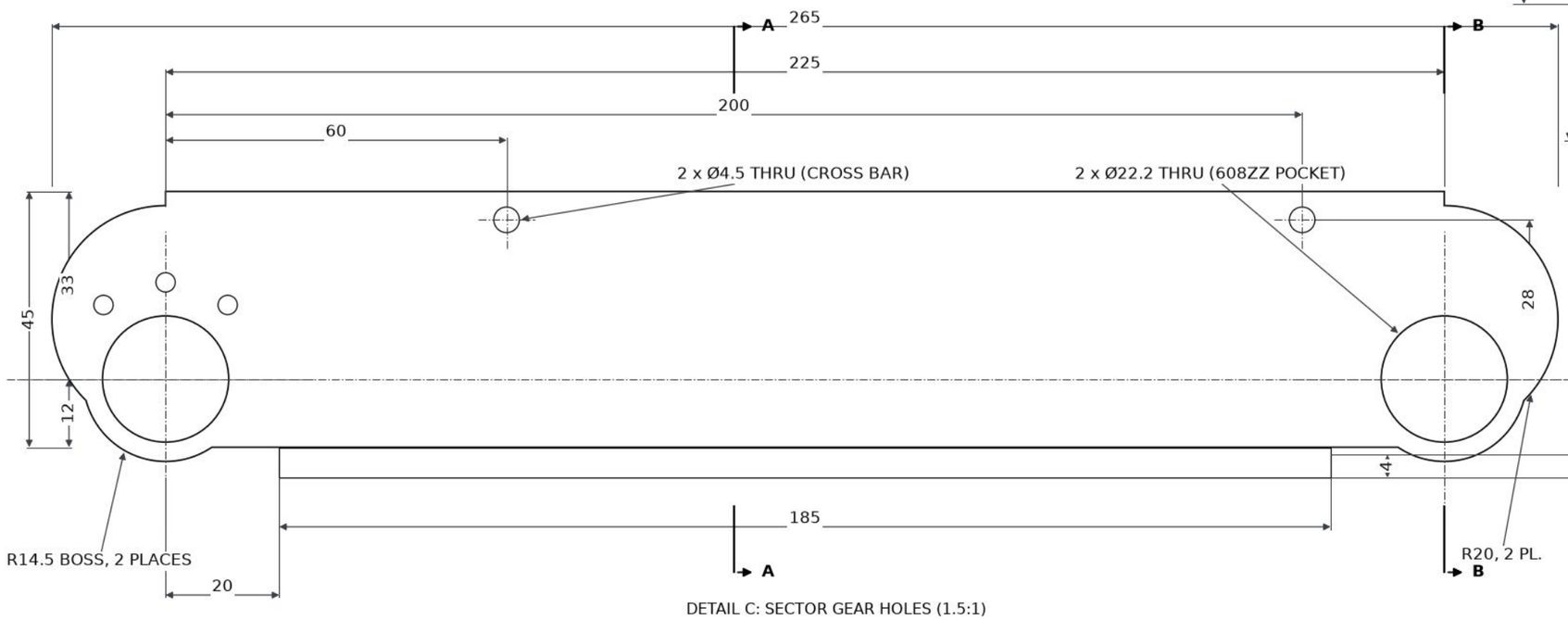
6 Notes

Print standing up as a ring, no supports, 100% infill, TPU 95A, on the Fab Lab UP300. Four tracks, about 32 g each.

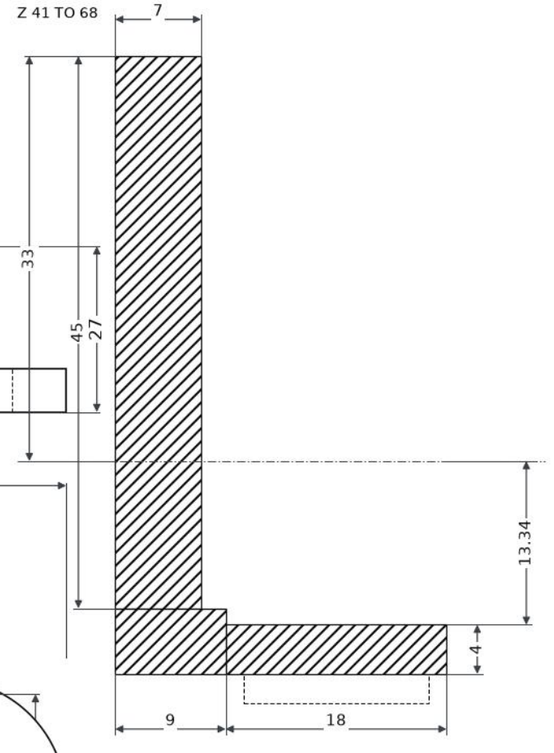
TOP VIEW



FRONT VIEW (FROM OUTBOARD)

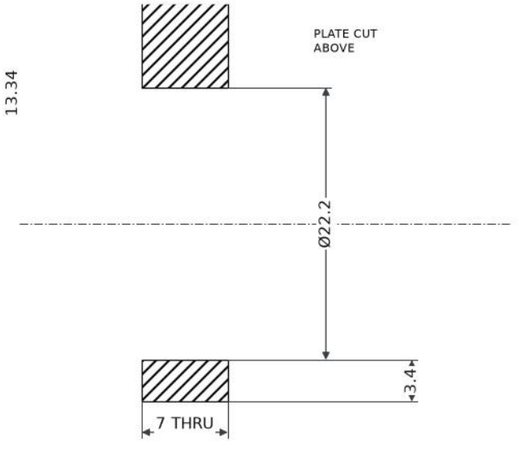


SECTION A-A (2:1)

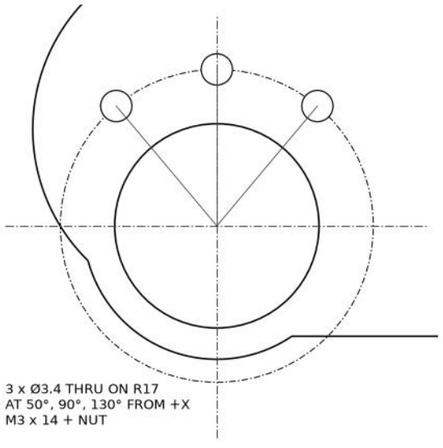


TRACK LOWER RUN (REF)

SECTION B-B (2:1)



DETAIL C: SECTOR GEAR HOLES (1.5:1)



3 x Ø3.4 THRU ON R17
AT 50°, 90°, 130° FROM +X
M3 x 14 + NUT

THIRD ANGLE

NOTES:
1. LEFT PLATE SHOWN; THE RIGHT ONE IS ITS MIRROR. FILES stl/side_plate_front_L.stl, _R.stl.
2. PRINT LYING ON THE PLATE FACE, 0.2 LAYERS, 4 WALLS, 30% INFILL, NO SUPPORTS.
3. BED 235 MM (OR 225 DIAGONAL); ON A SMALLER BED ASK FOR THE TWO-PIECE PLATE.
4. Z FROM THE VEHICLE CENTRE LINE: PLATE 41-48, WEB 41-50, RAIL 50-68
5. (RAIL CENTRED UNDER THE TRACK LANE 49-69). R20 ENDS ARE CENTRED 10.5 ABOVE THE AXLE.
6. POCKETS Ø22.2 THRU TAKE A 608ZZ (TS-07), LIGHT PRESS: MEASURE THE FIRST PRINT,
7. REAM IF TIGHT, GLUE IF LOOSE. THE RAIL TOP CARRIES THE LOWER RUN'S TOOTH TIPS.
8. X = 0 IS THE MIDDLE AXLE (HINGE); 3 x M3 SECTOR GEAR HOLES ON R17 ROUND IT, INBOARD FACE.

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:

SIDE PLATE, FRONT SEGMENT

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO. TS-02

QTY. 2 (1 L + 1 R)

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

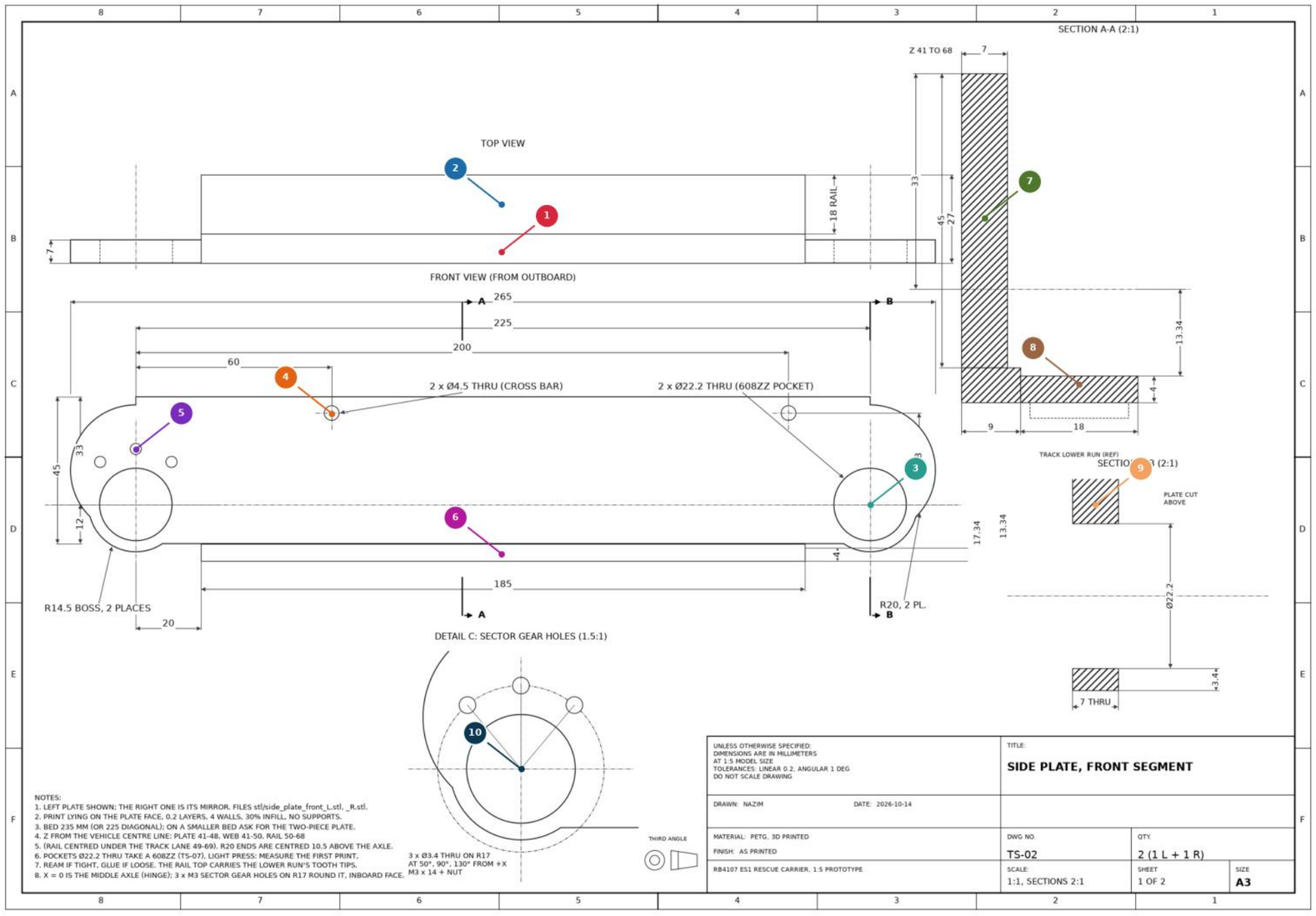
SCALE: 1:1, SECTIONS 2:1

SHEET 1 OF 2

SIZE **A3**

How to read TS-02: the front segment's side plate

Markers below sit on a reduced copy of sheet TS-02 (previous page, full size). Numbers match the key on the right.



1 Plate, seen from above

7 thick, 265 long, 45 tall. Printed flat on the bed in PETG. Left plate drawn; the right one is its mirror.

2 Slide rail, seen from above

An 18 wide, 4 thick strip 185 long, joined to the plate by a web. It sits under the track's lower run so the track cannot sag between the wheels.

3 Bearing pocket

Ø22.2 hole, one at each axle. A 608ZZ bearing is pressed in; the axle turns in it. The round boss round it keeps a 3.4 wall.

4 Cross bar hole

Ø4.5, two per plate at 60 and 200 from the axle, 28 above it. The M4 rods go through here.

5 Sector gear holes (front plate only)

Three Ø3.4 holes on R17 round the middle axle. The hinge sector gear bolts here with M3 x 14 and nuts, on the inboard face.

6 Rail from the side

The strip hanging below the plate between x 20 and 205. Its top is 13.34 below the axle centre: exactly where the track's tooth tips run.

7 Section A-A: plate

Cut through the rail at x 100 and looking along the plate: the 7 thick plate standing 45 tall.

8 Section A-A: rail and web

The rail (18 x 4) and the 9 web joining it to the plate, with the track's lower run dashed on top of it.

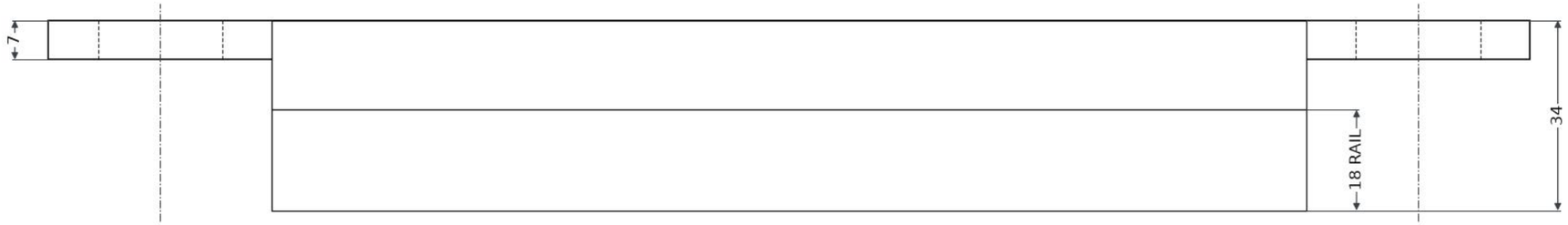
9 Section B-B: pocket

Cut through a bearing pocket: 7 thick, Ø22.2 straight through, 3.4 of wall round it.

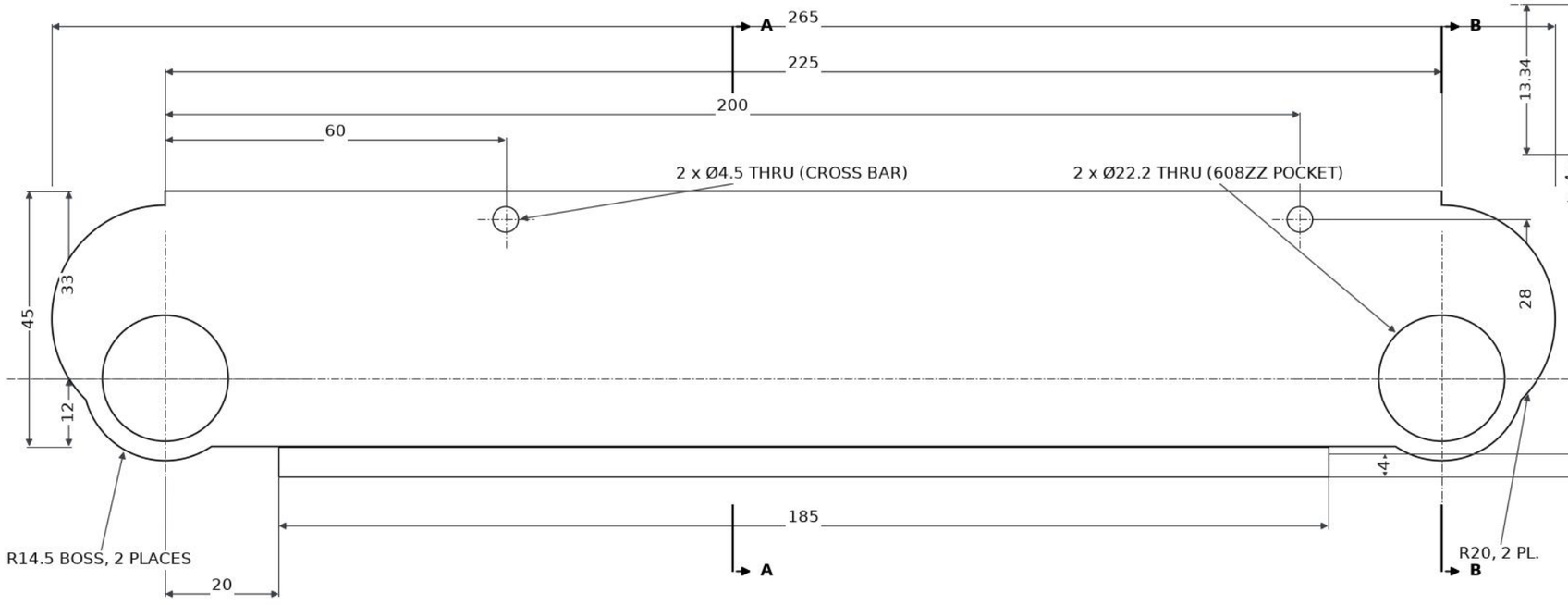
10 Detail C

The three sector gear holes at 50, 90 and 130 degrees from the forward direction, 1.5 times size.

TOP VIEW

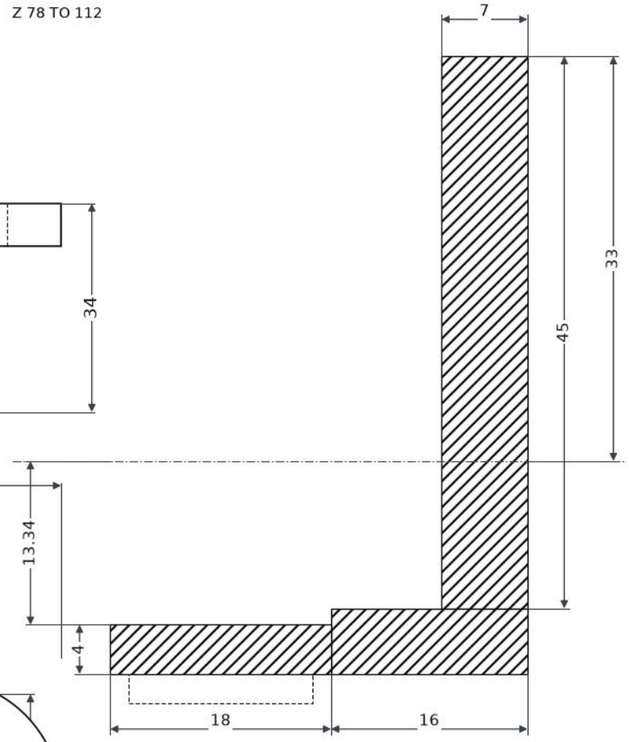


FRONT VIEW (FROM OUTBOARD)



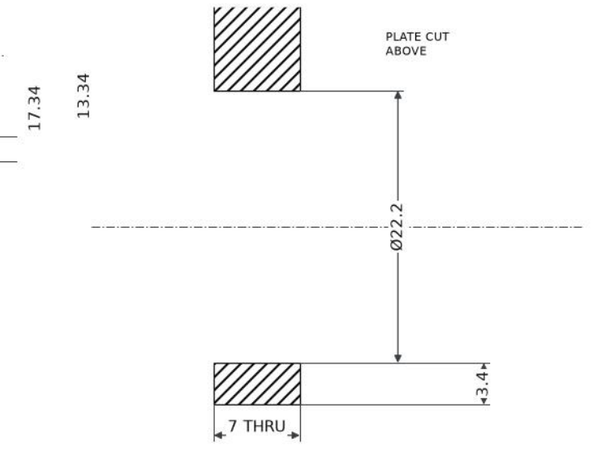
SECTION A-A (2:1)

Z 78 TO 112



TRACK LOWER RUN (REF)

SECTION B-B (2:1)



NOTES:
1. LEFT PLATE SHOWN; THE RIGHT ONE IS ITS MIRROR. FILES stl/side_plate_rear_L.stl, _R.stl.
2. PRINT LYING ON THE PLATE FACE, 0.2 LAYERS, 4 WALLS, 30% INFILL, NO SUPPORTS.
3. BED 235 MM (OR 225 DIAGONAL); ON A SMALLER BED ASK FOR THE TWO-PIECE PLATE.
4. Z FROM THE VEHICLE CENTRE LINE: PLATE 105-112, WEB 96-112, RAIL 78-96
5. (RAIL CENTRED UNDER THE TRACK LANE 77-97). R20 ENDS ARE CENTRED 10.5 ABOVE THE AXLE.
6. POCKETS Ø22.2 THRU TAKE A 608ZZ (TS-07), LIGHT PRESS: MEASURE THE FIRST PRINT,
7. REAM IF TIGHT, GLUE IF LOOSE. THE RAIL TOP CARRIES THE LOWER RUN'S TOOTH TIPS.
8. X = 0 IS THE REAR AXLE. THE BOX BOLTS TO THESE PLATES THROUGH THE Ø4.5 CROSS-BAR HOLES.

THIRD ANGLE



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
SIDE PLATE, REAR SEGMENT

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-03

QTY.: 2 (1 L + 1 R)

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

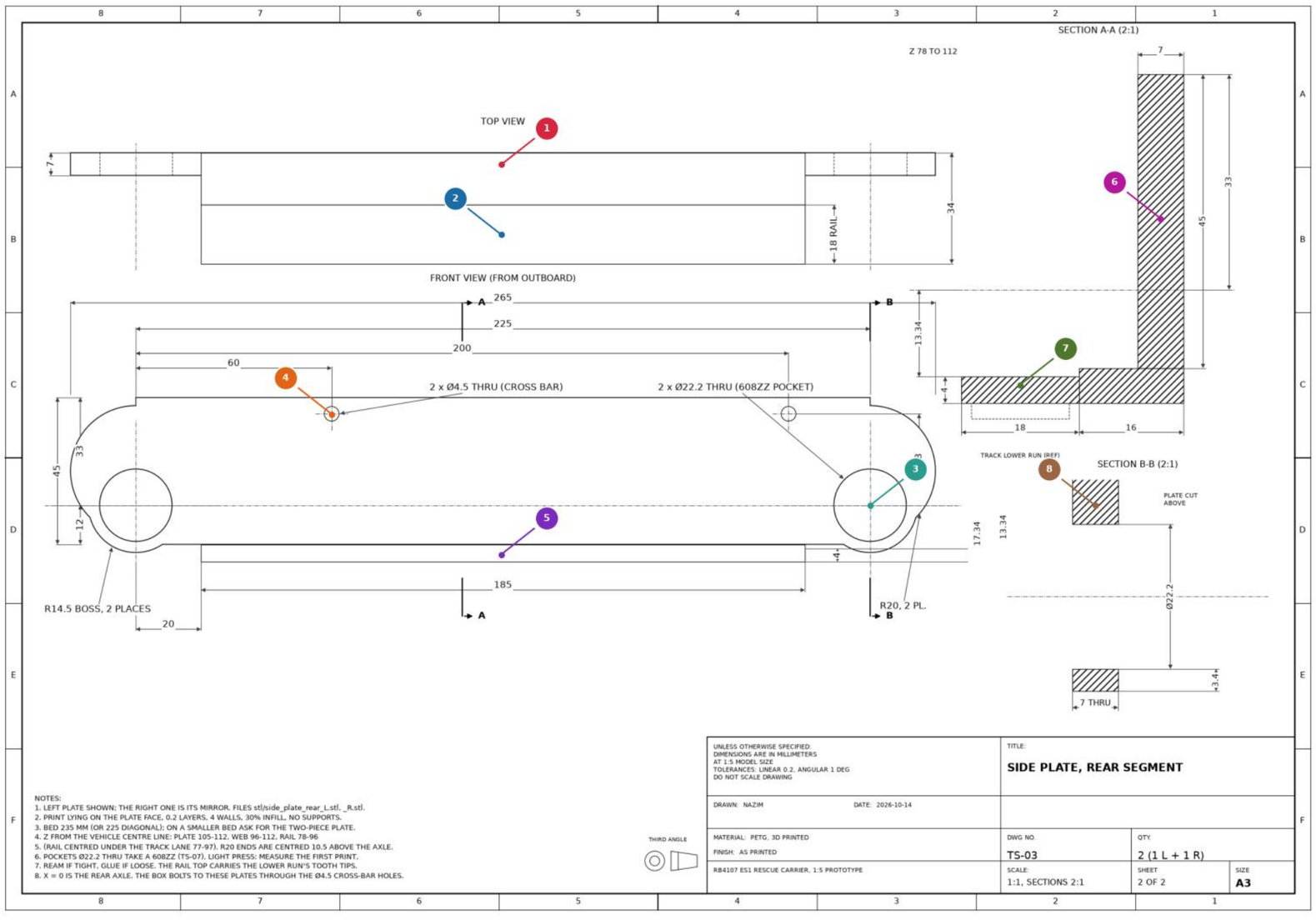
SCALE: 1:1, SECTIONS 2:1

SHEET: 2 OF 2

SIZE: **A3**

How to read TS-03: the rear segment's side plate

Markers below sit on a reduced copy of sheet TS-03 (previous page, full size). Numbers match the key on the right.



1 Plate, seen from above

Same plate as TS-02 but on the outside of the vehicle (z 105 to 112), so the rail is on its inboard face and it has no sector gear holes.

2 Slide rail, inboard

The rail sits under the rear track lane (z 78 to 96), joined to the plate by a 16 wide web that passes under the pulley hub.

3 Bearing pocket

Ø22.2, a 608ZZ is pressed in at each axle (rear axle and middle axle).

4 Cross bar hole

Ø4.5 at 60 and 200 from the rear axle, 28 up. The M4 rods through these also clamp the box: plate, spacer, box wall, box wall, spacer, plate (sheet TS-08).

5 Rail from the side

Hangs below the plate between x 20 and 205; its top carries the lower run's tooth tips.

6 Section A-A: plate

The 7 thick plate, 45 tall, now on the right of the section because it is outboard.

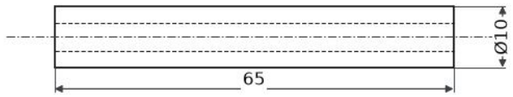
7 Section A-A: rail

18 x 4 rail with the track's lower run dashed on it, and the 16 web to the plate.

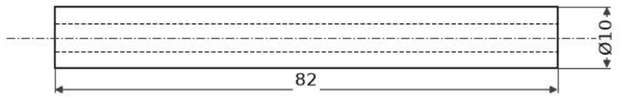
8 Section B-B: pocket

Cut through a pocket: Ø22.2 through the 7 plate, 3.4 wall.

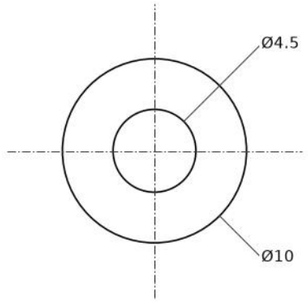
REAR SEGMENT SPACER, QTY 4 (2 PER ROD)



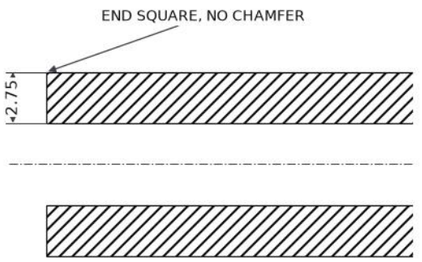
FRONT SEGMENT SPACER, QTY 2



END VIEW (3:1)



SECTION, ONE END (3:1)



NOTES:
1. FILES stl/spacer_front_x2.stl (82) AND stl/spacer_rear_x4.stl (65). PRINT STANDING ON END, 0.2 LAYERS, 3 WALLS, 20% INFILL.
2. FRONT: ONE 82 SPACER PER ROD BETWEEN THE FRONT PLATES (2 x 41). REAR: TWO 65 SPACERS PER ROD, ONE EACH SIDE BETWEEN THE BOX WALL (Z 40) AND THE REAR PLATE (Z 105).
3. THE NUTS OUTSIDE THE PLATES PULL PLATE - SPACER - BOX - SPACER - PLATE TOGETHER (SEE TS-08). AT 28 ABOVE THE AXLE THE SPACER CLEARS THE UPPER RUN'S CLEAT TIPS BY 3.3.
4. BORE 4.5 IS A CLEARANCE FOR THE M4 ROD. ENDS SQUARE: THEY SET THE PLATE SPACING.

THIRD ANGLE

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
AT 1:5 MODEL SIZE
TOLERANCES: LINEAR 0.2, ANGULAR 1 DEG
DO NOT SCALE DRAWING

TITLE:
CROSS BAR SPACER

DRAWN: NAZIM DATE: 2026-10-14

MATERIAL: PETG, 3D PRINTED
FINISH: AS PRINTED

DWG NO.: TS-05

QTY.: 2 x 82 + 4 x 65

RB4107 ES1 RESCUE CARRIER, 1:5 PROTOTYPE

SCALE: 1:1, DETAIL 3:1

SHEET: 1 OF 1

SIZE: **A3**

How to read TS-05: the cross bar spacers

Markers below sit on a reduced copy of sheet TS-05 (previous page, full size). Numbers match the key on the right.

1 Rear spacer, 65 long, 4 off

Two per rear rod, one each side between the box wall (z 40) and the rear plate (z 105). The box sits between them.

2 Front spacer, 82 long, 2 off

One per front rod, filling the whole gap between the two front plates (z -41 to 41).

3 End view

Ø10 outside, Ø4.5 bore: the M4 rod passes through with clearance.

4 Section

Just a tube, 2.75 wall, ends cut square because they set the plate spacing.

